Alkaline-Earth-Fluoride–LaF_3 Systems with Implications for Electroslag Melting

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Liquidus and solidus curves in the systems CaF_2-LaF_3, SrF_2-LaF_3, BaF_2-LaF_3, and MgF_2-LaF_3 under ⅓ atm helium pressure were determined by thermal analysis as an aid in the selection of suitable fluxes for electroslag melting of reactive metals. The systems CaF_2-LaF_3, SrF_2-LaF_3, and BaF_2-LaF_3 each contain (1) one eutectic, (2) no intermediate phases, and (3) two solid solutions whose lattice parameters vary linearly with composition. The liquidus curves in the systems SrF_2-LaF_3 and BaF_2-LaF_3 exhibit maxima. The system MgF_2-LaF_3 is of the simple eutectic type with no detectable terminal solid solutions. The smallest temperature interval observed between a fluoride-flux liquidus and the melting point of Ti occurs at the dystectic composition in the system SrF_2-LaF_3. Differences in density between metal and flux are more favorable in the system BaF_2-LaF_3 than in pure LaF_3. These observations suggest potential advantages of using fluxes from the systems studied for electroslag melting of reactive metals.

I. Introduction

ELECTROSLAG melting, a technique which complements vacuum-arc, induction, and electron-beam processes, has been applied extensively and successfully to steels and superalloys and with varying degrees of success to reactive and refractory metals.[1-4] It appears that the physicochemical properties of the fluxes are the most important parameters in establishing successful melting operations. Desirable flux characteristics for the electroslag process, which were enumerated previously,[1,2,4] include the requirements that: (1) The liquidus temperature of the flux should be slightly below that of the metal to be melted; (2) the density of the flux must be less than that of the metal at its melting point; and (3) the electrical conductivity of the flux should be as low as possible for economical operation (i.e. less than that of the metal but high enough to permit a stable melting operation). In the present work, four alkaline-earth-fluoride–LaF_3 systems which contain compositions that might be useful as fluxes were investigated by thermal analysis.

Group IIA fluorides may be subdivided, primarily on the basis of the radii of the cations, into two groups which in turn largely determine the coordination numbers of the ions and the crystal structures. The groups include those compounds with cationic radii* (in angstroms) which are (1) <1.0, i.e. BeF_2, 0.35 (0.34) and MgF_2, 0.80 (0.78); and (2) >1.0, i.e. CaF_2, 1.20 (1.06); SrF_2, 1.33 (1.27); BaF_2, 1.50 (1.43); and RaF_2, 1.56 (1.52). Compounds in both groups which have potential for application in the electroslag process were selected for study with LaF_3. The melting point of LaF_3 is $\approx$90°C higher than that of CaF_2, the flux most commonly used for electroslag melting of reactive metals.[1-3] In addition, LaF_3 is more stable than CaF_2 at the high temperatures involved in electroslag melting.

The systems CaF_2-LaF_3, SrF_2-LaF_3, BaF_2-LaF_3, and MgF_2-LaF_3 were studied because it was thought that they were most likely to form high-melting-point compounds, intermediate phases, and/or dystectics.† Such characteristics should decrease the temperature interval between flux liquidus temperatures and the melting point of reactive metals such as Ti and Zr and thus improve the surfaces of the melted metal. Relative high-temperature stabilities, the existence of a dystectic in an analogous YF_3 system,[9] and high-temperature compatibility with reactive metals were also considered in selecting these systems.

Literature data for these systems at high temperatures are scanty. Ippolitov and co-workers[10,11] presented results on the subsolidus phases in LaF_3-containing systems but gave no data on liquidus temperatures. Zagorskaya *et al.*[8] reported four eutectics and four dystectics in the CaF_2-LaF_3 system but did not specify the atmosphere used; their diagram may have resulted from excessive hydrolysis or nonequilibrium conditions. Ippolitov *et al.*,[12] however, obtained a diagram for this system in Ar similar to that derived in the present work, and Zhigarnovskii and Ippolitov[13] published a phase diagram for the system BaF_2-LaF_3 in 1.5 to 2 atm Ar. Hahn *et al.*,[14] who studied lattice parameters and room-temperature densities in fluorite solid-solution phases in several LaF_3-containing systems at 1200° and 1400°C, reported neither solidus nor liquidus temperatures.

II. Experimental Method

Reagent-grade fluorides served as starting materials in all the present work. This quality represents a compromise between that sufficient to provide accurate data and that expected to be used in commercial electroslag melting facilities. The fluorides were heated with excess $NH_4F{\cdot}HF$ under vacuum ($\approx 10^{-2}$ torr) to $\approx$900°C for several hours in a liner of an Mo-based alloy (0.02 C, 0.5 Ti, 0.08 Zr (wt%) balance Mo) contained in a graphite-resistor furnace. The total impurity content was typically 0.07 to 0.46 wt%. Twenty-gram mixtures were weighed in the correct proportions, ground, mixed to ensure homogeneity, and stored in desiccators.

Nuclear-grade graphite crucibles were used to contain the mixtures for thermal analysis. The determinations were conducted in a water-cooled graphite resistance furnace supplied with dc power. The emf values from an Nb-sheathed W3Re–W25Re thermocouple inserted into the sample were recorded as a function of time on calibrated adjustable-zero, adjustable-range millivolt potentiometric recorders to provide the necessary heating and cooling curves. The thermocouple was cali-

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*The first values given are the radii of Whittaker and Muntus (Ref. 5) for cationic coordination numbers of 4 and 6 for group (1) and of 8 for group (2). These values have been modified from those given by Shannon and Prewitt (Ref. 6). Values in parentheses are Goldschmidt radii (Ref. 7), which do not account for coordination numbers.

†The term dystectic, as used in this work, denotes a point of maximum temperature on the liquidus curve where the compositions of two equilibrium phases become identical and the system loses one degree of freedom. This definition follows previous usage (Ref. 8).

brated* frequently against the freezing points of oxygen-free Cu and Ni,† neither of which contained spectrographic impurities >0.3 ppm. No reactions between the thermocouple, sheath, sample, and crucible were observed. Before each run, the furnace was pumped to $\approx 10^{-2}$ torr and then backfilled with helium to ⅓ atm total pressure to approximate the conditions used for the electroslag melting of reactive metals in this laboratory. Samples were heated to ≈50° to 100°C above the liquidus temperatures and cooled. Heating and cooling rates were typically 30° and 25°C min^{-1}, respectively.

The initial changes in the slopes of the cooling curves identified the liquidus temperatures; solidus temperatures were determined from the heating curves when the solidus curves were relatively steep. The samples were observed visually with an optical pyrometer during heating and cooling to verify the solidus and liquidus determinations. The variance in the temperature measurements was estimated from several replicate runs on each mixture. After the thermal analysis runs, the samples were usually analyzed spectrographically for impurities, assayed for carbon and nitrogen, and observed by microscopy under reflected light. Phases were identified by powder X-ray diffraction methods occasionally supplemented by electron microprobe examination. X-ray lattice parameters were determined at room temperature on samples which had been held at constant temperature for 20 to 75 min and quenched by shutting off the furnace power. Such times are consistent with those used in electroslag melting practice.

The techniques described resulted in samples whose bulk compositions did not change significantly during thermal analysis, as is illustrated in Table I for mixtures in the system CaF_2-LaF_3. The average total weight loss of the CaF_2-LaF_3 samples during the runs was 1.8 wt% of the amount present.

*On the basis of the International Practical Temperature Scale of 1968 (Ref. 15).

†Grade I, Johnson Matthey & Co., New York, N. Y.

Table I. Bulk Compositions Before and After Thermal Analysis for Mixtures in the System CaF_2-LaF_3

Nominal composition (wt% LaF_3)	Calculated* composition (wt% LaF_3)	Difference (wt% LaF_3)
20.0	17.3	−2.7
29.0	29.1	+0.1
38.6	37.2	−1.4
52.0	52.0	0
62.6	64.7	+2.1
71.5	74.2	+2.7

*Calculated values were derived from La analyses by X-ray emission techniques. Analytical errors are estimated to be ±1% of the amount present.

Comparable differences were observed in the other systems. Supplemental DTA was conducted under conditions described previously[16] on selected compositions in each of the systems studied.

III. Results

Liquidus and solidus curves determined from the heating and cooling curves for the systems CaF_2-LaF_3, SrF_2-LaF_3, BaF_2-LaF_3, and MgF_2-LaF_3 are presented in Fig. 1, in which uncertainty estimates are indicated by the length of the vertical line for each mixture. Subsolidus relations were inferred from selected X-ray lattice parameter measurements and from microscopic observations.

No solid solutions were detected by X-ray in the eutectic MgF_2-LaF_3 system (Table II), although minute amounts of both solid solutions must be present. Each mixture in this system exhibited a eutectic structure after thermal analysis;

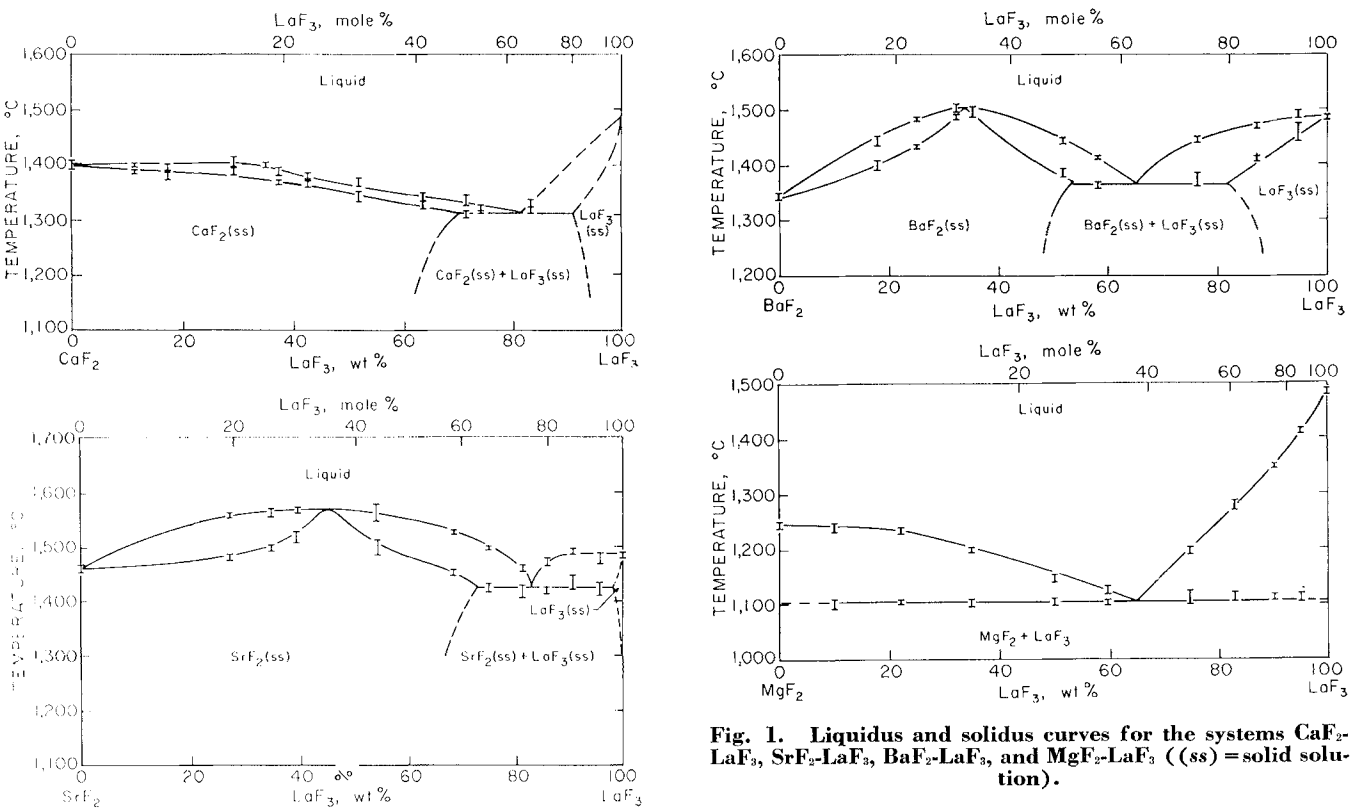


Fig. 1. Liquidus and solidus curves for the systems CaF_2-LaF_3, SrF_2-LaF_3, BaF_2-LaF_3, and MgF_2-LaF_3 ((ss) = solid solution).

Table II. X-Ray Lattice Parameters for Phases in the System MgF_2-LaF_3

Composition (wt% MgF_2)	a_0 (Å)	c_0 (Å)
	MgF_2	
100	4.623±0.002	3.052±0.002
90	4.623±0.002	3.050±0.002
65	4.623±0.002	3.050±0.002
	LaF_3	
25	7.185±0.005	7.355±0.005
17	7.187±0.005	7.347±0.005
10	7.187±0.001	7.355±0.004
5	7.188±0.003	7.352±0.007
0	7.185±0.005	7.354±0.005

the textures of representative compositions are shown in Fig. 2. The eutectic composition contains 35±1 wt% MgF_2 at 1105°C.

The other systems had similar characteristics, including (1) two solid solutions of the fluorite (CaF_2) and tysonite (LaF_3) structures separated by a solid immiscibility gap and (2) a eutectic. The SrF_2-LaF_3 and BaF_2-LaF_3 systems each contain a maximum on the liquidus curve whose temperature is ≈110°C and 160°C, respectively, above the melting point of the alkaline-earth fluoride. This maximum is located at compositions where the primary phase is the fluorite solid solution; the LaF_3 solid solution is much more limited than the fluorite solid solution in each of these systems. No compounds or other intermediate phases were detected in any of the systems studied.

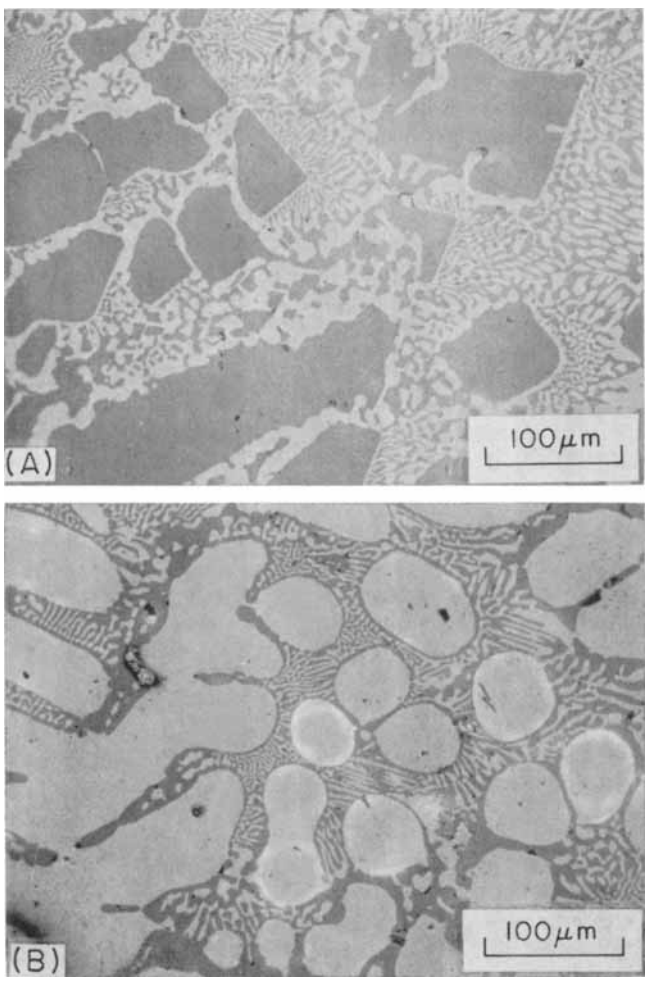


Fig. 2. Compositions in the system MgF_2-LaF_3 after thermal analysis. (*A*) MgF_2 grains (*dark areas*) in eutectic matrix, 65 wt% MgF_2 and (*B*) LaF_3 grains (*light areas*) in eutectic matrix, 25 wt% MgF_2.

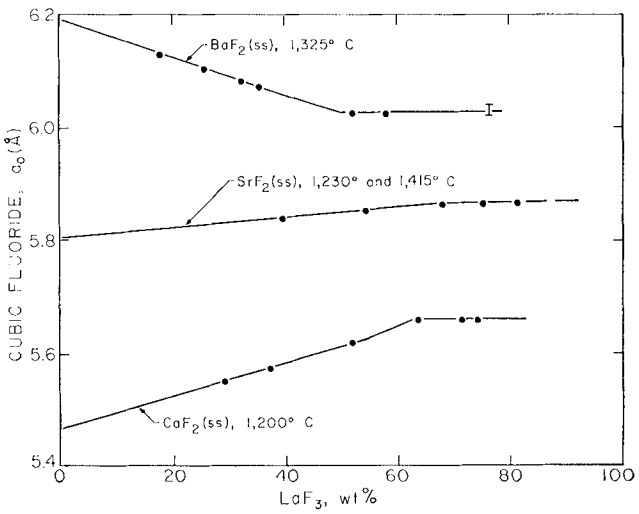


Fig. 3. Variation of lattice parameters as functions of LaF_3 content in the systems CaF_2-LaF_3, SrF_2-LaF_3, and BaF_2-LaF_3.

The X-ray data are shown as functions of composition in Fig. 3 for the systems in which extensive fluorite solid solutions are present. The lattice parameters for these cubic fluorite interstitial solid solutions change linearly with composition. The lattice parameters of mixtures in the CaF_2-LaF_3 system exhibit the greatest increase as a function of increasing LaF_3 content, whereas those in the SrF_2-LaF_3 system are more independent of composition. On the other hand, the fluorite lattice parameter in the BaF_2-LaF_3 system decreases with increasing LaF_3 content. These observations are consistent with the relative cationic radii (see * footnote, p. 130; LaF_3=1.26 (Refs. 5 and 6) (1.22, Ref. 7) for an La^{3+} coordination number of 8) when the effect of interstitial fluoride ions is considered. On the basis of cationic radii alone, an increase in the lattice parameter of the fluorite solid solution with increasing LaF_3 content would be expected for the system CaF_2-LaF_3. The observed increase (Fig. 3) is also attributed to the additional interstitial F^- ions derived from the substitution of LaF_3 for CaF_2. In the SrF_2-LaF_3 system, a decrease in lattice size is expected on the basis of cationic radii alone. The slight increase observed (Fig. 3) probably results from the contribution of the interstitial F^- ions. No lattice parameter increase was noted in the BaF_2-LaF_3 system because the relatively large difference in cationic radii overrides the opposing superimposed effect of the interstitial F^- ions. The solid solutions based on LaF_3 are less extensive than the fluorite solid solutions in these three systems; pertinent lattice parameters are given in Table III.

IV. Discussion

Since no solid solutions, compounds, or intermediate phases were observed in the MgF_2-LaF_3 system, the high-temperature thermodynamic properties of the liquid phase can be estimated.[17] The heat of fusion for MgF_2 was taken as 13,900± 200 cal mol^{-1} (Ref. 18), and that for LaF_3 was estimated from the liquidus data in Fig. 1 and phase diagram theory as 14,500 cal mol^{-1}. At the melting point of Ti (1948°±7°K), the calculated activities of each component in the composition range where it is the primary phase show nearly ideal relations. The activities of the components over the remainder of the system were calculated from a binary integration of the Gibbs-

Table III. X-Ray Lattice Parameters for the Tysonite Phase (LaF_3)*

Composition (wt% LaF_3)	a_0 (Å)	c_0 (Å)
	All systems	
100	7.185±0.005	7.354±0.005
	CaF_2-LaF_3 (1200°C)	
83.5	7.114±0.005	7.328±0.005
74.2	7.117±0.005	7.323±0.008
71.5	≈7.12	≈7.26
	SrF_2-LaF_3 (1230° and 1415°C)	
95.41	7.18±0.01	7.35±0.02
90.68	7.18±0.01	7.36±0.01
81.4	7.17±0.01	7.37±0.01
75.3	7.17±0.01	7.37±0.01
	BaF_2-LaF_3 (1325°C)	
94.7	7.205±0.005	7.39 ±0.01
87.5	7.249±0.007	7.417±0.007
76.5	7.269±0.005	7.428±0.005
58.1	7.24 ±0.01	7.41 ±0.01

*Determined at room temperature on samples quenched from temperature(s) indicated.

Table IV. Densities of Molten Fluorides at or Slightly Above Melting Points of Reactive Metals*

Metal	Ti	Zr	V
Temp. (°K)	1948	2108	2203–2213
Density (g-cm^{-3}) of:			
Metal	4.11	6.06	5.734
CaF_2	2.417	2.358	2.319
SrF_2	3.33	3.23	3.126
BaF_2	3.824	<3.75	<3.75
MgF_2	2.213	2.133	<2.125
LaF_3	4.462	4.357	4.29

*Data were taken from Refs. 20 and 21. Temperature given is that at which the density of the metal was determined; the fluoride densities were interpolated or extrapolated from data in the literature.

Duhem equation using the α-function.[19] These activities exhibit slightly negative deviations from ideality. The activity vs composition data permit calculation of the free energy of mixing (ΔG^m) in kcal mol^{-1} according to the equation:

$$\Delta G^m = \frac{RT(N_{MgF_2} \ln a_{MgF_2} + N_{LaF_3} \ln a_{LaF_3})}{1000} \quad (1)$$

where R is the gas constant (1.987 cal deg.$^{-1}$ mol^{-1}) and T the melting point of Ti. When these data are plotted vs composition, ΔG^m is an almost symmetrical function of composition, indicating simple solution behavior for this system at temperatures of interest in electroslag melting.

In the absence of specific determinations of component activities (e.g. by vapor pressure measurements or emf cells) very little can be said definitely concerning the high-temperature activities in the liquid phase of the systems CaF_2-LaF_3, SrF_2-LaF_3, and BaF_2-LaF_3. In each system, the heats of mixing of the solid-solution phases are greater than those for the liquid phase, both of which are greater than zero.

Although no high-melting-point compounds were detected in any of the systems studied, significant maxima on the liquidus curves were observed in two of the systems. These dystectics are believed to result from the aforementioned substitution of interstitial fluoride ions into the fluorite solid solution. This substitution introduces defects into the crystal structure, resulting in a less-than-normal entropy of fusion because of the added disorder. This effect increases the ratio of the heat of fusion to the entropy of fusion and thus increases the liquidus temperature.[9]

In the system SrF_2-LaF_3, the dystectic temperature is only ≈100°C lower than the melting point of Ti. This interval is the smallest between the liquidus temperature of a flux and the melting point of a reactive metal observed in any fluoride system thus far; hence, this dystectic represents a desirable flux composition for the melting of Ti. The liquidus maximum in the BaF_2-LaF_3 system occurs at a temperature slightly higher than the LaF_3 melting point. Molten LaF_3, however, is denser at the melting point of Ti than Ti itself, thus making LaF_3 unsuitable for the electroslag melting of Ti.[2] On the other hand, the flux density of the BaF_2-LaF_3 dystectic is expected, on the basis of a linear interpolation between the fluoride components in each system as a first approximation, to be lower than that of Ti at its melting point. The approximate compositions (wt% LaF_3) where flux densities at the melting point of Ti become larger than the density of the metal are 48, 71, 83, and 84 for the BaF_2-LaF_3, SrF_2-LaF_3, CaF_2-LaF_3, and MgF_2-LaF_3 systems, respectively. Therefore, flux compositions which contain more than these amounts of LaF_3 are unsuitable for electroslag melting of Ti. All flux compositions in these systems are less dense than Zr and V at temperatures above their respective melting points, as indicated in Table IV.

No solid-state transitions were observed in any of the component fluorides in the four systems. The estimated limits of solid solubility determined from the present data agree to within 3 mol% for both solid solutions, compared with those observed by previous investigators.[10,11,13]

Data on high-temperature electrical conductivities, which are considered important flux parameters in the electroslag process, are sparse for these systems. Bacon *et al.*[22] determined the electrical conductivity of molten CaF_2-LaF_3 fluxes; with LaF_3 additions up to 20 wt%, the specific conductivity at 1500° to 1600°C decreased significantly. These workers concluded that this behavior prevents arcing by causing insufficient power density and melt rate in the electroslag process.

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Kinetics of Recrystallization and Grain Growth in Cu_2O

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The kinetics of recrystallization and grain growth in Cu_2O prepared by oxidation of Cu at 1030°C were studied. A critical strain of 8% at 700°C is necessary to cause recrystallization during subsequent annealing at 1020°C. Grain-growth kinetics fit the expression $D^3-D_0^3=kt$ with an activation energy of 83±2 kcal/mol.

I. Introduction

PROLONGED annealing at 1110°C of flat plates of Cu_2O produced by oxidizing Cu in air at 1020° to 1065°C promotes grain growth.[1] If the grains produced are large enough, single-crystal specimens can be made from them; such specimens have been used to study the plastic deformation of Cu_2O.[2] The present investigation of the kinetics of recrystallization and grain growth was undertaken in an attempt to improve this technique.

II. Experimental Methods

Cuprous oxide was prepared by controlled oxidation of squares of OHFC copper 2 by 25 by 25 mm. These squares were mechanically polished and suspended for 48 h from a Pt hook in a vertical furnace maintained at 1030°±5°C. This procedure produced specimens of Cu_2O 3 by 27 by 27 mm. The microstructures of these specimens, which were revealed by mechanical polishing and etching in 35% HNO_3, consisted of equiaxed grains. Grain size was measured by the intercept method[3,4] on a plane parallel to the specimen surface.

A thicker Cu plate would be needed to produce thicker squares of the oxide, and the oxidation would require several weeks (the temperature must remain below 1065°C to avoid formation of the Cu_2O-Cu eutectic). Therefore, specimens of Cu_2O 9 by 27 by 27 mm were prepared by bonding together three of the thinner squares either by diffusion during a 60-h anneal at 1050°C or by application of a compressive strain of 0.2 at 760°C and initiation of recrystallization with a subsequent 5-h anneal at 1050°C.

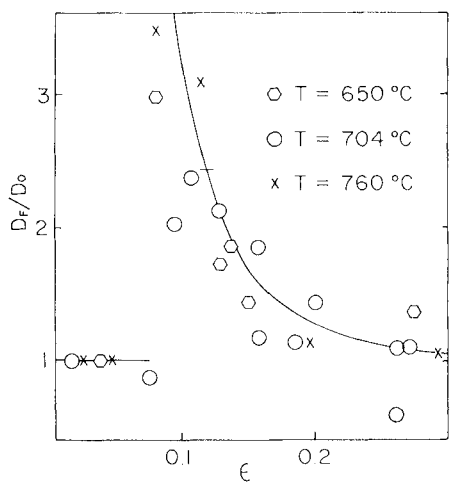


Fig. 1. Variation of the ratio of final grain size to initial grain size (D_F/D_0) with compressive strain (ε) after 4-h recrystallization anneal at 1020°C. T=deformation temperature; values of D_0 were between 0.63 and 1.10 mm.

Specimens 9 by 10 by 10 mm for use in compression tests were cut from these blocks. Compression tests were performed on a testing machine§ at 650° to 760°C using an initial strain rate of 0.22/min. After it was deformed, the specimen was cut into halves; one half served as a control and the other was annealed for 4 h at 1020°C to promote recrystallization.

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